Intake Artifacts

INTAKE_FORM_CONFIG.v1.json

```
{
  "artifact_type": "INTAKE_FORM_CONFIG",
  "form_version": "1.0.0",
  "page_order": [
    "page_0_routing",
    "page_1_project_basics_inputs",
    "page_2_intent_brand",
    "page_3_design_direction",
    "page_4_functional_spec",
    "page_5_data_model",
    "page_6_auth",
    "page_7_integrations",
    "page_8_nfr",
    "page_9_category_specific",
    "page_10_final_confirmations"
  ],
  "routing_axes": [
    "skill_level",
    "category",
    "type_preset",
    "build_target",
    "audience_context",
    "autofill"
  ],
  "routing_page": {
    "id": "page_0_routing",
    "title": "Routing (Matrix Selector)",
    "fields": [
      { "path": "routing.skill_level", "ui": "radio", "required": true },
      { "path": "routing.category", "ui": "dropdown", "required": true },
      { "path": "routing.type_preset", "ui": "cards", "required": true, "filters": ["routing.category"] },
      { "path": "routing.build_target", "ui": "radio", "required": true },
      { "path": "routing.audience_context", "ui": "radio", "required": true },
      { "path": "routing.autofill", "ui": "toggle", "required": true, "default": false }
    ]
  },
  "pages": [
    {
      "id": "page_1_project_basics_inputs",
      "title": "Project Basics + Inputs",
      "modules": [
        {
          "id": "project_identity",
          "fields": [
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        { "path": "project.project_name", "ui": "text", "required": true },
        { "path": "project.project_overview", "ui": "textarea_lg", "required": true },
        { "path": "project.project_oneliner", "ui": "text", "required": false },
        { "path": "project.core_experience_paragraph", "ui": "textarea", "required": false }
      ]
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    {
      "id": "source_materials",
      "fields": [
        { "path": "inputs.zip_upload", "ui": "file_zip", "required": false },
        { "path": "inputs.reference_links", "ui": "repeatable_url_label", "required": false }
      ]
    },
    {
      "id": "existing_project_context",
      "when": { "eq": ["routing.build_target", "existing"] },
      "fields": [
        { "path": "existing.existing_repo_link", "ui": "url", "required": true },
        { "path": "existing.current_state_summary", "ui": "textarea", "required": true },
        { "path": "existing.must_not_change", "ui": "list_builder", "required": false },
        { "path": "existing.known_issues", "ui": "repeatable_title_desc", "required": false }
      ]
    },
    {
      "id": "delivery_preference",
      "fields": [
        { "path": "delivery.desired_scope", "ui": "radio", "required": true },
        { "path": "delivery.priority_bias", "ui": "radio", "required": true }
      ]
    }
  ]
},
{
  "id": "page_2_intent_brand",
  "title": "Product Intent + Brand Ideology",
  "modules": [
    {
      "id": "intent_core",
      "fields": [
        { "path": "intent.problem_statement", "ui": "textarea", "required": true },
        { "path": "intent.why_now", "ui": "textarea", "required": false },
        { "path": "intent.alternatives", "ui": "repeatable_name_link_missing", "required": false },
        { "path": "intent.audience_description", "ui": "textarea", "required": true },
        { "path": "intent.primary_user_label", "ui": "text", "required": true },
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          { "path": "intent.secondary_users", "ui": "repeatable_text", "required": false },
          { "path": "intent.primary_goals", "ui": "repeatable_text", "required": true },
          { "path": "intent.success_definition", "ui": "textarea", "required": true },
          { "path": "intent.success_metrics", "ui": "repeatable_metric_target_time", "required": false
},
          { "path": "intent.out_of_scope", "ui": "repeatable_text", "required": true }
        ]
      },
      {
        "id": "consumer_brand_blocks",
        "when": { "eq": ["routing.audience_context", "consumer"] },
        "fields": [
          { "path": "brand.stands_for", "ui": "textarea", "required": true },
          { "path": "brand.change_created", "ui": "textarea", "required": false },
          { "path": "brand.core_values", "ui": "chips_custom", "required": false },
          { "path": "brand.decision_beliefs", "ui": "chips_custom", "required": false },
          { "path": "brand.refusal_guardrails", "ui": "textarea", "required": false },
          { "path": "brand.psychographics", "ui": "textarea", "required": false },
          { "path": "brand.user_cares_about", "ui": "textarea", "required": false },
          { "path": "brand.user_fears_avoids", "ui": "textarea", "required": false },
          { "path": "brand.aligned_communities", "ui": "repeatable_text", "required": false },
          { "path": "brand.status_signal", "ui": "dropdown", "required": false },
          { "path": "brand.brand_promise", "ui": "text", "required": true },
          { "path": "brand.positioning_statement", "ui": "textarea", "required": false },
          { "path": "brand.unique_advantage", "ui": "textarea", "required": false },
          { "path": "brand.voice_adjectives", "ui": "chips_custom", "required": true },
          { "path": "brand.tone_boundaries", "ui": "textarea", "required": false },
          { "path": "brand.example_phrases", "ui": "repeatable_text", "required": false },
          { "path": "brand.content_maturity", "ui": "radio", "required": false },
          { "path": "brand.first_open_emotion", "ui": "textarea", "required": false },
          { "path": "brand.post_success_emotion", "ui": "textarea", "required": false },
          { "path": "brand.avoid_emotions", "ui": "textarea", "required": false }
        ]
      }
    ]
  },
  {
    "id": "page_3_design_direction",
    "title": "Design Direction",
    "modules": [
      {
        "id": "design_core",
        "fields": [
          {
```

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            "path": "design.style_adjectives",
            "ui": "chips_custom",
            "required": false,
            "required_when": { "eq": ["routing.audience_context", "consumer"] }
          },
          { "path": "design.visual_preset", "ui": "dropdown", "required": false },
          { "path": "design.avoid_list", "ui": "list_builder", "required": false },
          { "path": "design.logo_file", "ui": "file", "required": false },
          { "path": "design.brand_colors", "ui": "repeatable_name_hex", "required": false },
          { "path": "design.fonts", "ui": "repeatable_name_usecase", "required": false },
          { "path": "design.brand_references", "ui": "repeatable_url_notes", "required": false },
          { "path": "design.ui_density", "ui": "radio", "required": false },
          { "path": "design.nav_preference", "ui": "dropdown", "required": false },
          { "path": "design.component_vibe", "ui": "chips_custom", "required": false },
          { "path": "design.motion_preference", "ui": "radio", "required": false },
          { "path": "design.a11y_expectations", "ui": "checklist", "required": false },
          { "path": "design.a11y_notes", "ui": "textarea", "required": false },
          { "path": "design.content_presentation", "ui": "radio", "required": false },
          { "path": "design.showcase_screens", "ui": "repeatable_showcase", "required": false }
        ]
      }
    ]
  },
  {
    "id": "page_4_functional_spec",
    "title": "Functional Spec",
    "modules": [
      {
        "id": "features",
        "fields": [
          { "path": "spec.must_have_features", "ui": "repeatable_name_desc", "required": true },
          { "path": "spec.nice_to_have_features", "ui": "repeatable_name_desc", "required": false
},
          { "path": "spec.future_features", "ui": "repeatable_name_desc", "required": false },
          { "path": "spec.feature_priority_rank", "ui": "drag_drop_names", "required": true }
        ]
      },
      {
        "id": "roles_permissions",
        "fields": [
          { "path": "spec.roles", "ui": "repeatable_role", "required": true },
          { "path": "spec.role_permissions", "ui": "repeatable_permissions", "required": true }
        ]
      },
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    {
      "id": "workflows",
      "fields": [
        { "path": "spec.workflows", "ui": "repeatable_workflow", "required": true },
        { "path": "spec.edge_workflows", "ui": "repeatable_workflow", "required": false }
      ]
    },
    {
      "id": "business_rules",
      "fields": [
        { "path": "spec.business_rules_enabled", "ui": "toggle", "required": false },
        { "path": "spec.must_always_rules", "ui": "repeatable_text", "required": false },
        { "path": "spec.must_never_rules", "ui": "repeatable_text", "required": false },
        { "path": "spec.validation_rules", "ui": "repeatable_field_rule", "required": false },
        { "path": "spec.lifecycle_rules", "ui": "repeatable_text", "required": false }
      ]
    }
  ]
},
{
  "id": "page_5_data_model",
  "title": "Data Model",
  "modules": [
    {
      "id": "data_gate",
      "fields": [{ "path": "data.enabled", "ui": "radio_yes_no", "required": true }]
    },
    {
      "id": "data_details",
      "when": { "eq": ["data.enabled", true] },
      "fields": [{ "path": "data.objects", "ui": "repeatable_data_object", "required": true }]
    }
  ]
},
{
  "id": "page_6_auth",
  "title": "Auth & Authorization",
  "modules": [
    {
      "id": "auth_gate",
      "fields": [{ "path": "auth.required", "ui": "radio_yes_no", "required": true }]
    },
    {
      "id": "auth_details",
```

```
        "when": { "eq": ["auth.required", true] },
        "fields": [
          { "path": "auth.methods", "ui": "multi_select", "required": true },
          { "path": "auth.account_lifecycle", "ui": "multi_select", "required": false },
          { "path": "auth.two_fa_required", "ui": "radio_yes_no", "required": false },
          { "path": "auth.session_rules", "ui": "multi_select", "required": false },
          { "path": "auth.authorization_model", "ui": "dropdown", "required": false },
          { "path": "auth.approval_flows_needed", "ui": "radio_yes_no", "required": false },
          { "path": "auth.approval_flows_description", "ui": "textarea", "required": false }
        ]
      }
    ]
  },
  {
    "id": "page_7_integrations",
    "title": "Integrations",
    "modules": [
      {
        "id": "integrations_gate",
        "fields": [{ "path": "integrations.enabled", "ui": "radio_yes_no", "required": true }]
      },
      {
        "id": "integrations_details",
        "when": { "eq": ["integrations.enabled", true] },
        "fields": [{ "path": "integrations.items", "ui": "repeatable_integration", "required": true }]
      }
    ]
  },
  {
    "id": "page_8_nfr",
    "title": "Non-Functional Requirements (NFR)",
    "modules": [
      {
        "id": "nfr_core",
        "fields": [
          { "path": "nfr.performance_targets", "ui": "repeatable_metric_target", "required": false },
          { "path": "nfr.scale_assumptions", "ui": "object_users_concurrency_notes", "required":
false },
          { "path": "nfr.reliability_expectation", "ui": "dropdown", "required": false },
          { "path": "nfr.offline_required", "ui": "radio_yes_no", "required": false },
          { "path": "nfr.compliance_flags", "ui": "multi_select", "required": false }
        ]
      }
    ]
```

```
    },
    {
      "id": "page_9_category_specific",
      "title": "Category/Type Specific Page",
      "modules": [
        { "id": "cat_application", "when": { "eq": ["routing.category", "application"] }, "fields": [{ "path": "app_ui", "ui": "application_ui_block", "required": true }] },
        { "id": "cat_service", "when": { "eq": ["routing.category", "service"] }, "fields": [{ "path": "service", "ui": "service_block", "required": true }] },
        { "id": "cat_infra", "when": { "eq": ["routing.category", "infrastructure"] }, "fields": [{ "path": "infra", "ui": "infra_block", "required": true }] },
        { "id": "cat_sdk", "when": { "eq": ["routing.category", "library_sdk"] }, "fields": [{ "path": "sdk", "ui": "sdk_block", "required": true }] }
      ]
    },
    {
      "id": "page_10_final_confirmations",
      "title": "Final Verification + Confirmations",
      "modules": [
        {
          "id": "final_core",
          "fields": [
            { "path": "final.definition_of_done", "ui": "textarea", "required": true },
            { "path": "final.must_pass_checks", "ui": "multi_select_custom", "required": false },
            { "path": "final.acceptance_criteria", "ui": "repeatable_text", "required": false },
            { "path": "final.rejection_conditions", "ui": "textarea", "required": false },
            { "path": "final.confirm_priorities", "ui": "checkbox", "required": true },
            { "path": "final.confirm_out_of_scope", "ui": "checkbox", "required": true },
            { "path": "final.confirm_constraints", "ui": "checkbox", "required": true }
          ]
        }
      ]
    }
  ]
}
```

INTAKE_SCHEMA.v1.json

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{
  "artifact_type": "INTAKE_SCHEMA",
  "schema_version": "1.0.0",
  "type": "object",
  "required": ["schema_version", "form_version", "routing", "project", "delivery", "intent", "spec",
"final"],
  "properties": {
    "schema_version": { "type": "string" },
    "form_version": { "type": "string" },

    "routing": {
      "type": "object",
      "required": ["skill_level", "category", "type_preset", "build_target", "audience_context",
"autofill"],
      "properties": {
        "skill_level": { "$ref": "#/enums/SkillLevel" },
        "category": { "$ref": "#/enums/ProjectCategory" },
        "type_preset": { "$ref": "#/enums/TypePreset" },
        "build_target": { "$ref": "#/enums/BuildTarget" },
        "audience_context": { "$ref": "#/enums/AudienceContext" },
        "autofill": { "type": "boolean" }
      }
    },

    "project": {
      "type": "object",
      "required": ["project_name", "project_overview"],
      "properties": {
        "project_name": { "type": "string", "minLength": 1, "maxLength": 120 },
        "project_overview": { "type": "string", "minLength": 20, "maxLength": 4000 },
        "project_oneliner": { "type": "string", "maxLength": 180 },
        "core_experience_paragraph": { "type": "string", "maxLength": 1200 }
      }
    },

    "inputs": {
      "type": "object",
      "properties": {
        "zip_upload": { "$ref": "#/types/fileRef" },
        "reference_links": {
          "type": "array",
          "items": { "type": "object", "required": ["url"], "properties": { "url": { "type": "string", "format":
"uri" }, "label": { "type": "string", "maxLength": 80 } } }
        }
```

```
      }
    },

    "existing": {
      "type": "object",
      "properties": {
        "existing_repo_link": { "type": "string", "format": "uri" },
        "current_state_summary": { "type": "string", "minLength": 20, "maxLength": 3000 },
        "must_not_change": { "type": "array", "items": { "type": "string", "maxLength": 200 } },
        "known_issues": { "type": "array", "items": { "type": "object", "required": ["title"], "properties":
{ "title": { "type": "string", "maxLength": 120 }, "description": { "type": "string", "maxLength": 800 }
} } }
      }
    },

    "delivery": {
      "type": "object",
      "required": ["desired_scope", "priority_bias"],
      "properties": {
        "desired_scope": { "$ref": "#/enums/DesiredScope" },
        "priority_bias": { "$ref": "#/enums/PriorityBias" }
      }
    },

    "intent": {
      "type": "object",
      "required": ["problem_statement", "audience_description", "primary_user_label",
"primary_goals", "success_definition", "out_of_scope"],
      "properties": {
        "problem_statement": { "type": "string", "minLength": 20, "maxLength": 3000 },
        "why_now": { "type": "string", "maxLength": 2000 },
        "alternatives": {
          "type": "array",
          "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string",
"maxLength": 120 }, "link": { "type": "string", "format": "uri" }, "whats_missing": { "type": "string",
"maxLength": 500 } } }
        },
        "audience_description": { "type": "string", "minLength": 20, "maxLength": 2000 },
        "primary_user_label": { "type": "string", "minLength": 1, "maxLength": 80 },
        "secondary_users": { "type": "array", "items": { "type": "string", "maxLength": 80 } },
        "primary_goals": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 140
} },
        "success_definition": { "type": "string", "minLength": 10, "maxLength": 2000 },
        "success_metrics": {
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"type": "array",
"items": { "type": "object", "required": ["metric"], "properties": { "metric": { "type": "string", "maxLength": 140 }, "target": { "type": "string", "maxLength": 120 }, "timeframe": { "type": "string", "maxLength": 120 } } }
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"out_of_scope": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 160 } }
}
},

"brand": {
"type": "object",
"properties": {
"stands_for": { "type": "string", "maxLength": 2000 },
"change_created": { "type": "string", "maxLength": 2000 },
"core_values": { "type": "array", "items": { "type": "string", "maxLength": 60 } },
"decision_beliefs": { "type": "array", "items": { "type": "string", "maxLength": 80 } },
"refusal_guardrails": { "type": "string", "maxLength": 2000 },
"psychographics": { "type": "string", "maxLength": 2000 },
"user_cares_about": { "type": "string", "maxLength": 1200 },
"user_fears_avoids": { "type": "string", "maxLength": 1200 },
"aligned_communities": { "type": "array", "items": { "type": "string", "maxLength": 100 } },
"status_signal": { "$ref": "#/enums/StatusSignal" },
"brand_promise": { "type": "string", "maxLength": 160 },
"positioning_statement": { "type": "string", "maxLength": 400 },
"unique_advantage": { "type": "string", "maxLength": 400 },
"voice_adjectives": { "type": "array", "items": { "type": "string", "maxLength": 40 } },
"tone_boundaries": { "type": "string", "maxLength": 800 },
"example_phrases": { "type": "array", "items": { "type": "string", "maxLength": 140 } },
"content_maturity": { "$ref": "#/enums/ContentMaturity" },
"first_open_emotion": { "type": "string", "maxLength": 300 },
"post_success_emotion": { "type": "string", "maxLength": 300 },
"avoid_emotions": { "type": "string", "maxLength": 300 }
}
},

"design": {
"type": "object",
"properties": {
"style_adjectives": { "type": "array", "items": { "type": "string", "maxLength": 40 } },
"visual_preset": { "$ref": "#/enums/VisualPreset" },
"avoid_list": { "type": "array", "items": { "type": "string", "maxLength": 80 } },
"logo_file": { "$ref": "#/types/fileRef" },

"brand_colors": { "type": "array", "items": { "type": "object", "required": ["hex"], "properties": { "name": { "type": "string", "maxLength": 40 }, "hex": { "type": "string", "maxLength": 9 } } } },
"fonts": { "type": "array", "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string", "maxLength": 80 }, "use_case": { "$ref": "#/enums/FontUseCase" } } } },
"brand_references": { "type": "array", "items": { "type": "object", "required": ["url"], "properties": { "url": { "type": "string", "format": "uri" }, "notes": { "type": "string", "maxLength": 200 } } } },
"ui_density": { "$ref": "#/enums/UIDensity" },
"nav_preference": { "$ref": "#/enums/NavPreference" },
"component_vibe": { "type": "array", "items": { "type": "string", "maxLength": 40 } },
"motion_preference": { "$ref": "#/enums/MotionPreference" },
"a11y_expectations": { "type": "array", "items": { "type": "string", "maxLength": 60 } },
"a11y_notes": { "type": "string", "maxLength": 800 },
"content_presentation": { "$ref": "#/enums/ContentPresentation" },
"showcase_screens": {
"type": "array",
"items": {
"type": "object",
"required": ["name"],
"properties": {
"name": { "type": "string", "maxLength": 80 },
"visual_priority": { "$ref": "#/enums/ShowcasePriority" },
"notes": { "type": "string", "maxLength": 400 },
"ref_url": { "type": "string", "format": "uri" },
"ref_file": { "$ref": "#/types/fileRef" }
}
}
}
}
},

"spec": {
"type": "object",
"required": ["must_have_features", "feature_priority_rank", "roles", "role_permissions", "workflows"],
"properties": {
"must_have_features": { "type": "array", "minItems": 1, "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string", "maxLength": 120 }, "description": { "type": "string", "maxLength": 800 } } } },
"nice_to_have_features": { "type": "array", "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string", "maxLength": 120 }, "description": { "type": "string", "maxLength": 800 } } } },

"future_features": { "type": "array", "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string", "maxLength": 120 }, "description": { "type": "string", "maxLength": 800 } } } },
"feature_priority_rank": { "type": "array", "items": { "type": "string", "maxLength": 120 } },

"roles": { "type": "array", "minItems": 1, "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string", "maxLength": 60 }, "description": { "type": "string", "maxLength": 400 }, "primary_goal": { "type": "string", "maxLength": 200 } } } },
"role_permissions": {
"type": "array",
"items": {
"type": "object",
"required": ["role_name", "allowed_capabilities"],
"properties": {
"role_name": { "type": "string", "maxLength": 60 },
"allowed_capabilities": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 80 } },
"restricted_actions": { "type": "array", "items": { "type": "string", "maxLength": 80 } },
"approval_required_actions": { "type": "array", "items": { "type": "string", "maxLength": 80 } }
}
}
},

"workflows": {
"type": "array",
"minItems": 3,
"items": {
"type": "object",
"required": ["name", "actor_role", "steps", "success_outcome"],
"properties": {
"name": { "type": "string", "maxLength": 120 },
"actor_role": { "type": "string", "maxLength": 60 },
"steps": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 200 } },
"success_outcome": { "type": "string", "maxLength": 200 },
"failure_states": { "type": "string", "maxLength": 400 },
"priority": { "$ref": "#/enums/WorkflowPriority" }
}
}
},
"edge_workflows": { "type": "array", "items": { "type": "object" } },

"business_rules_enabled": { "type": "boolean" },
"must_always_rules": { "type": "array", "items": { "type": "string", "maxLength": 200 } },

"must_never_rules": { "type": "array", "items": { "type": "string", "maxLength": 200 } },
"validation_rules": { "type": "array", "items": { "type": "object", "required": ["rule"], "properties": { "fields": { "type": "string", "maxLength": 200 }, "rule": { "type": "string", "maxLength": 200 } } } },
"lifecycle_rules": { "type": "array", "items": { "type": "string", "maxLength": 200 } }
}
},

"data": { "type": "object", "properties": { "enabled": { "type": "boolean" }, "objects": { "type": "array", "items": { "type": "object", "required": ["name"], "properties": { "name": { "type": "string", "maxLength": 80 } } } } } } },
"auth": { "type": "object", "properties": { "required": { "type": "boolean" } } },
"integrations": { "type": "object", "properties": { "enabled": { "type": "boolean" } } },
"nfr": { "type": "object" },

"app_ui": { "type": "object" },
"service": { "type": "object" },
"infra": { "type": "object" },
"sdk": { "type": "object" },

"final": {
"type": "object",
"required": ["definition_of_done", "confirm_priorities", "confirm_out_of_scope", "confirm_constraints"],
"properties": {
"definition_of_done": { "type": "string", "minLength": 10, "maxLength": 2000 },
"must_pass_checks": { "type": "array", "items": { "type": "string", "maxLength": 120 } },
"acceptance_criteria": { "type": "array", "items": { "type": "string", "maxLength": 200 } },
"rejection_conditions": { "type": "string", "maxLength": 1200 },
"confirm_priorities": { "type": "boolean" },
"confirm_out_of_scope": { "type": "boolean" },
"confirm_constraints": { "type": "boolean" }
}
}
},

"types": {
"fileRef": {
"type": "object",
"required": ["file_id", "name"],
"properties": {
"file_id": { "type": "string", "minLength": 6, "maxLength": 80 },
"name": { "type": "string", "maxLength": 200 },
"mime": { "type": "string", "maxLength": 80 },

```
        "size_bytes": { "type": "integer", "minimum": 0 }
      }
    }
  },

  "enums": {
    "SkillLevel": { "type": "string", "enum": ["beginner", "intermediate", "expert"] },
    "ProjectCategory": { "type": "string", "enum": ["application", "service", "infrastructure",
"library_sdk"] },
    "TypePreset": { "type": "string", "enum": ["TBD"] },
    "BuildTarget": { "type": "string", "enum": ["new", "existing"] },
    "AudienceContext": { "type": "string", "enum": ["consumer", "internal", "developer"] },

    "DesiredScope": { "type": "string", "enum": ["mvp", "full", "unsure"] },
    "PriorityBias": { "type": "string", "enum": ["speed", "balanced", "quality"] },

    "StatusSignal": { "type": "string", "enum": ["TBD"] },
    "ContentMaturity": { "type": "string", "enum": ["TBD"] },

    "VisualPreset": { "type": "string", "enum": ["TBD"] },
    "FontUseCase": { "type": "string", "enum": ["heading", "body", "mono", "brand", "other"] },
    "UIDensity": { "type": "string", "enum": ["compact", "balanced", "spacious"] },
    "NavPreference": { "type": "string", "enum": ["tabs", "sidebar", "top_nav", "hybrid", "unsure"] },
    "MotionPreference": { "type": "string", "enum": ["none", "subtle", "expressive"] },
    "ContentPresentation": { "type": "string", "enum": ["text_first", "visual_first", "balanced"] },
    "ShowcasePriority": { "type": "string", "enum": ["low", "medium", "high"] },

    "WorkflowPriority": { "type": "string", "enum": ["p0", "p1", "p2", "unsure"] }
  }
}
```

INTAKE_RULESET.v1.json

```
{
  "artifact_type": "INTAKE_RULESET",
  "ruleset_version": "1.0.0",
  "rule_format": {
    "required_fields": ["rule_id", "description", "when", "then", "error_code", "pointer_paths",
"severity"]
  },
  "rules": [
    {
      "rule_id": "INT3-ROUTING-01",
      "description": "type_preset must be valid for the selected category",
      "when": { "always": true },
      "then": [{ "op": "preset_valid_for_category", "args": ["routing.category", "routing.type_preset"]
}],
      "error_code": "INVALID_ENUM_FOR_CONTEXT",
      "pointer_paths": ["routing.category", "routing.type_preset"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-EXIST-01",
      "description": "Existing projects must provide repo link and current state summary",
      "when": { "eq": ["routing.build_target", "existing"] },
      "then": [
        { "op": "required", "path": "existing.existing_repo_link" },
        { "op": "format", "path": "existing.existing_repo_link", "format": "uri" },
        { "op": "required", "path": "existing.current_state_summary" }
      ],
      "error_code": "DEPENDENCY_MISSING",
      "pointer_paths": ["existing.existing_repo_link", "existing.current_state_summary"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-CONS-01",
      "description": "Consumer projects require core brand minimums",
      "when": { "eq": ["routing.audience_context", "consumer"] },
      "then": [
        { "op": "required", "path": "brand.stands_for" },
        { "op": "required", "path": "brand.brand_promise" },
        { "op": "min_items", "path": "brand.voice_adjectives", "min": 1 }
      ],
      "error_code": "DEPENDENCY_MISSING",
      "pointer_paths": ["brand.stands_for", "brand.brand_promise", "brand.voice_adjectives"],
      "severity": "error"
    },
```

```
{
  "rule_id": "INT3-CONS-02",
  "description": "Consumer projects require at least one style adjective",
  "when": { "eq": ["routing.audience_context", "consumer"] },
  "then": [{ "op": "min_items", "path": "design.style_adjectives", "min": 1 }],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["design.style_adjectives"],
  "severity": "error"
},

{
  "rule_id": "INT3-DATA-00",
  "description": "Data gate must be answered",
  "when": { "always": true },
  "then": [{ "op": "required", "path": "data.enabled" }],
  "error_code": "REQUIRED",
  "pointer_paths": ["data.enabled"],
  "severity": "error"
},
{
  "rule_id": "INT3-DATA-01",
  "description": "If data is enabled, at least one data object must be provided",
  "when": { "eq": ["data.enabled", true] },
  "then": [{ "op": "min_items", "path": "data.objects", "min": 1 }],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["data.objects"],
  "severity": "error"
},

{
  "rule_id": "INT3-AUTH-00",
  "description": "Auth gate must be answered",
  "when": { "always": true },
  "then": [{ "op": "required", "path": "auth.required" }],
  "error_code": "REQUIRED",
  "pointer_paths": ["auth.required"],
  "severity": "error"
},
{
  "rule_id": "INT3-AUTH-01",
  "description": "If auth is required, at least one auth method must be selected",
  "when": { "eq": ["auth.required", true] },
  "then": [{ "op": "min_items", "path": "auth.methods", "min": 1 }],
  "error_code": "MIN_ITEMS",
```

```
  "pointer_paths": ["auth.methods"],
  "severity": "error"
},
{
  "rule_id": "INT3-AUTH-02",
  "description": "If approval flows are needed, approval flow description is required",
  "when": { "eq": ["auth.approval_flows_needed", true] },
  "then": [{ "op": "required", "path": "auth.approval_flows_description" }],
  "error_code": "DEPENDENCY_MISSING",
  "pointer_paths": ["auth.approval_flows_description"],
  "severity": "error"
},

{
  "rule_id": "INT3-INTEG-00",
  "description": "Integrations gate must be answered",
  "when": { "always": true },
  "then": [{ "op": "required", "path": "integrations.enabled" }],
  "error_code": "REQUIRED",
  "pointer_paths": ["integrations.enabled"],
  "severity": "error"
},
{
  "rule_id": "INT3-INTEG-01",
  "description": "If integrations are enabled, at least one integration item must be provided",
  "when": { "eq": ["integrations.enabled", true] },
  "then": [{ "op": "min_items", "path": "integrations.items", "min": 1 }],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["integrations.items"],
  "severity": "error"
},

{
  "rule_id": "INT3-REF-01",
  "description": "Role names must be unique",
  "when": { "exists": "spec.roles" },
  "then": [{ "op": "unique", "path": "spec.roles[].name" }],
  "error_code": "DUPLICATE_VALUE",
  "pointer_paths": ["spec.roles"],
  "severity": "error"
},
{
  "rule_id": "INT3-REF-02",
  "description": "Role permissions must reference valid role names",
```

"when": { "exists": "spec.role_permissions" },
"then": [{ "op": "ref_exists", "path": "spec.role_permissions[].role_name", "in": "spec.roles[].name" }],
"error_code": "INVALID_REFERENCE",
"pointer_paths": ["spec.role_permissions", "spec.roles"],
"severity": "error"
},
{
"rule_id": "INT3-REF-03",
"description": "Workflows must reference valid actor roles",
"when": { "exists": "spec.workflows" },
"then": [{ "op": "ref_exists", "path": "spec.workflows[].actor_role", "in": "spec.roles[].name" }],
"error_code": "INVALID_REFERENCE",
"pointer_paths": ["spec.workflows", "spec.roles"],
"severity": "error"
},
{
"rule_id": "INT3-REF-04",
"description": "Feature priority rank must reference must-have features by name",
"when": { "exists": "spec.feature_priority_rank" },
"then": [{ "op": "ref_exists", "path": "spec.feature_priority_rank[]", "in": "spec.must_have_features[].name" }],
"error_code": "INVALID_REFERENCE",
"pointer_paths": ["spec.feature_priority_rank", "spec.must_have_features"],
"severity": "error"
},
{
"rule_id": "INT3-REF-05",
"description": "Each workflow must have at least one step",
"when": { "exists": "spec.workflows" },
"then": [{ "op": "min_items_each", "path": "spec.workflows[].steps", "min": 1 }],
"error_code": "MIN_ITEMS",
"pointer_paths": ["spec.workflows"],
"severity": "error"
},

{
"rule_id": "INT3-RULES-01",
"description": "If business rules are enabled, must_always_rules must have at least 1 entry",
"when": { "eq": ["spec.business_rules_enabled", true] },
"then": [{ "op": "min_items", "path": "spec.must_always_rules", "min": 1 }],
"error_code": "MIN_ITEMS",
"pointer_paths": ["spec.must_always_rules"],
"severity": "error"

```
    },

    {
      "rule_id": "INT3-FMT-01",
      "description": "Zip uploads must be .zip",
      "when": { "exists": "inputs.zip_upload" },
      "then": [{ "op": "file_ext", "path": "inputs.zip_upload.name", "ext": ".zip" }],
      "error_code": "INVALID_FILETYPE",
      "pointer_paths": ["inputs.zip_upload"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-FMT-02",
      "description": "Reference links must be valid URLs",
      "when": { "exists": "inputs.reference_links" },
      "then": [{ "op": "format_each", "path": "inputs.reference_links[].url", "format": "uri" }],
      "error_code": "INVALID_URL",
      "pointer_paths": ["inputs.reference_links"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-FMT-03",
      "description": "Brand color hex values must be valid hex",
      "when": { "exists": "design.brand_colors" },
      "then": [{ "op": "regex_each", "path": "design.brand_colors[].hex", "pattern":
"^#([A-Fa-f0-9]{6}|[A-Fa-f0-9]{8})$" }],
      "error_code": "INVALID_FORMAT",
      "pointer_paths": ["design.brand_colors"],
      "severity": "error"
    },

    {
      "rule_id": "INT3-SKILL-01",
      "description": "Beginner must-have features minimum",
      "when": { "eq": ["routing.skill_level", "beginner"] },
      "then": [{ "op": "min_items", "path": "spec.must_have_features", "min": 1 }],
      "error_code": "MIN_ITEMS",
      "pointer_paths": ["spec.must_have_features"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-SKILL-02",
      "description": "Intermediate must-have features minimum",
      "when": { "eq": ["routing.skill_level", "intermediate"] },
```

```
  "then": [{ "op": "min_items", "path": "spec.must_have_features", "min": 3 }],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["spec.must_have_features"],
  "severity": "error"
},
{
  "rule_id": "INT3-SKILL-03",
  "description": "Expert must-have features minimum",
  "when": { "eq": ["routing.skill_level", "expert"] },
  "then": [{ "op": "min_items", "path": "spec.must_have_features", "min": 5 }],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["spec.must_have_features"],
  "severity": "error"
},
{
  "rule_id": "INT3-SKILL-04",
  "description": "Beginner/Intermediate workflows minimum",
  "when": { "in": ["routing.skill_level", ["beginner", "intermediate"]] },
  "then": [{ "op": "min_items", "path": "spec.workflows", "min": 3 }],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["spec.workflows"],
  "severity": "error"
},
{
  "rule_id": "INT3-SKILL-05",
  "description": "Expert workflow depth alternative",
  "when": { "eq": ["routing.skill_level", "expert"] },
  "then": [
    { "op": "either", "any": [
      { "op": "min_items", "path": "spec.workflows", "min": 5 },
      { "op": "all", "all": [
        { "op": "min_items", "path": "spec.workflows", "min": 3 },
        { "op": "min_items", "path": "spec.edge_workflows", "min": 2 }
      ] }
    ] }
  ],
  "error_code": "MIN_ITEMS",
  "pointer_paths": ["spec.workflows", "spec.edge_workflows"],
  "severity": "error"
},
{
  "rule_id": "INT3-SKILL-06",
  "description": "Beginner out-of-scope minimum",
  "when": { "eq": ["routing.skill_level", "beginner"] },
```

```
      "then": [{ "op": "min_items", "path": "intent.out_of_scope", "min": 1 }],
      "error_code": "MIN_ITEMS",
      "pointer_paths": ["intent.out_of_scope"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-SKILL-07",
      "description": "Intermediate out-of-scope minimum",
      "when": { "eq": ["routing.skill_level", "intermediate"] },
      "then": [{ "op": "min_items", "path": "intent.out_of_scope", "min": 2 }],
      "error_code": "MIN_ITEMS",
      "pointer_paths": ["intent.out_of_scope"],
      "severity": "error"
    },
    {
      "rule_id": "INT3-SKILL-08",
      "description": "Expert out-of-scope minimum",
      "when": { "eq": ["routing.skill_level", "expert"] },
      "then": [{ "op": "min_items", "path": "intent.out_of_scope", "min": 3 }],
      "error_code": "MIN_ITEMS",
      "pointer_paths": ["intent.out_of_scope"],
      "severity": "error"
    },

    {
      "rule_id": "INT3-FINAL-01",
      "description": "Final confirmations must be true",
      "when": { "always": true },
      "then": [
        { "op": "equals", "path": "final.confirm_priorities", "value": true },
        { "op": "equals", "path": "final.confirm_out_of_scope", "value": true },
        { "op": "equals", "path": "final.confirm_constraints", "value": true }
      ],
      "error_code": "REQUIRED",
      "pointer_paths": ["final.confirm_priorities", "final.confirm_out_of_scope",
"final.confirm_constraints"],
      "severity": "error"
    }
  ]
}
```

ENUM_REGISTRY.v1.json

```
{
  "artifact_type": "ENUM_REGISTRY",
  "enum_registry_version": "1.0.0",

  "enums": {
    "SkillLevel": ["beginner", "intermediate", "expert"],
    "ProjectCategory": ["application", "service", "infrastructure", "library_sdk"],
    "BuildTarget": ["new", "existing"],
    "AudienceContext": ["consumer", "internal", "developer"],

    "DesiredScope": ["mvp", "full", "unsure"],
    "PriorityBias": ["speed", "balanced", "quality"],

    "WorkflowPriority": ["p0", "p1", "p2", "unsure"],
    "FontUseCase": ["heading", "body", "mono", "brand", "other"],
    "UIDensity": ["compact", "balanced", "spacious"],
    "NavPreference": ["tabs", "sidebar", "top_nav", "hybrid", "unsure"],
    "MotionPreference": ["none", "subtle", "expressive"],
    "ContentPresentation": ["text_first", "visual_first", "balanced"],
    "ShowcasePriority": ["low", "medium", "high"],

    "StatusSignal": ["value", "premium", "luxury", "street", "minimal", "bold", "classic", "innovative",
"heritage"],
    "ContentMaturity": ["all_ages", "teen", "mature", "explicit"],
    "VisualPreset": ["clean_modern", "luxury_minimal", "bold_playful", "dark_premium",
"corporate", "retro", "editorial", "techno"]
  },

  "type_preset": {
    "values": [
      "web_app_public",
      "web_app_internal",
      "mobile_app",
      "desktop_app",
      "api_service",
      "event_driven_service",
      "batch_job",
      "infra_platform",
      "infra_pipeline",
      "sdk_client",
      "sdk_framework"
    ],
    "allowed_by_category": {
      "application": ["web_app_public", "web_app_internal", "mobile_app", "desktop_app"],
```

"service": ["api_service", "event_driven_service", "batch_job"],
"infrastructure": ["infra_platform", "infra_pipeline"],
"library_sdk": ["sdk_client", "sdk_framework"]
}
}
}

VALIDATION_REPORT_FORMAT.v1.jso
n

```
{
  "artifact_type": "VALIDATION_REPORT_FORMAT",
  "report_version": "1.0.0",
  "required_fields": [
    "schema_version",
    "form_version",
    "ruleset_version",
    "routing_snapshot",
    "result",
    "errors",
    "warnings",
    "included_modules"
  ],
  "shape": {
    "schema_version": "string",
    "form_version": "string",
    "ruleset_version": "string",
    "routing_snapshot": "object",
    "included_modules": "array<string>",
    "result": { "status": "pass|fail", "error_count": "number", "warning_count": "number" },
    "errors": "array<RuleHit>",
    "warnings": "array<RuleHit>"
  },
  "RuleHit": {
    "rule_id": "string",
    "error_code": "string",
    "severity": "error|warning",
    "message": "string",
    "pointer_paths": "array<string>",
    "details": "object",
    "evidence": {
      "received_value": "any",
      "expected": "any"
    }
  }
}
```

Canonical Artifacts

CANONICAL_SPEC_SCHEMA.v1.json

```
{
  "artifact_type": "CANONICAL_SPEC_SCHEMA",
  "spec_schema_version": "1.0.0",
  "description": "Authoritative schema for Canonical Spec + SPEC_INDEX registry.",
  "required": ["spec_schema_version", "project_id", "routing_snapshot", "spec_index"],
  "properties": {
    "spec_schema_version": { "type": "string" },
    "project_id": { "type": "string", "minLength": 3, "maxLength": 80 },
    "routing_snapshot": { "type": "object" },

    "meta": {
      "type": "object",
      "properties": {
        "created_at": { "type": "string", "format": "date-time" },
        "created_by": { "type": "string" },
        "source_intake": { "type": "object", "properties": { "schema_version": { "type": "string" },
"form_version": { "type": "string" } } },
        "notes": { "type": "string", "maxLength": 2000 }
      }
    },

    "spec_index": {
      "type": "object",
      "required": [
        "screen",
        "endpoint",
        "entity",
        "business_rule",
        "component",
        "acceptance_criterion",
        "test_case",
        "flow"
      ],
      "properties": {
        "screen": { "$ref": "#/defs/IndexList_Screen" },
        "endpoint": { "$ref": "#/defs/IndexList_Endpoint" },
        "entity": { "$ref": "#/defs/IndexList_Entity" },
        "business_rule": { "$ref": "#/defs/IndexList_BusinessRule" },
        "component": { "$ref": "#/defs/IndexList_Component" },
        "acceptance_criterion": { "$ref": "#/defs/IndexList_AcceptanceCriterion" },
        "test_case": { "$ref": "#/defs/IndexList_TestCase" },
        "flow": { "$ref": "#/defs/IndexList_Flow" }
      }
    },
```

```
    "reference_rules_version": { "type": "string" }
  },

  "defs": {
    "Ref": {
      "type": "string",
      "pattern": "^[a-z_]+:[A-Z]{3,6}-[0-9]{4}$"
    },

    "BaseItem": {
      "type": "object",
      "required": ["id", "title", "status"],
      "properties": {
        "id": { "$ref": "#/defs/Ref" },
        "title": { "type": "string", "minLength": 1, "maxLength": 160 },
        "status": { "type": "string", "enum": ["draft", "locked"] },
        "tags": { "type": "array", "items": { "type": "string", "maxLength": 40 } },
        "owner": { "type": "string", "maxLength": 60 },
        "notes": { "type": "string", "maxLength": 4000 },
        "refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
      }
    },

    "IndexList_Screen": { "type": "array", "items": { "$ref": "#/defs/Screen" } },
    "IndexList_Endpoint": { "type": "array", "items": { "$ref": "#/defs/Endpoint" } },
    "IndexList_Entity": { "type": "array", "items": { "$ref": "#/defs/Entity" } },
    "IndexList_BusinessRule": { "type": "array", "items": { "$ref": "#/defs/BusinessRule" } },
    "IndexList_Component": { "type": "array", "items": { "$ref": "#/defs/Component" } },
    "IndexList_AcceptanceCriterion": { "type": "array", "items": { "$ref":
"#/defs/AcceptanceCriterion" } },
    "IndexList_TestCase": { "type": "array", "items": { "$ref": "#/defs/TestCase" } },
    "IndexList_Flow": { "type": "array", "items": { "$ref": "#/defs/Flow" } },

    "Screen": {
      "allOf": [{ "$ref": "#/defs/BaseItem" }],
      "required": ["purpose", "actors", "states", "actions"],
      "properties": {
        "purpose": { "type": "string", "maxLength": 600 },
        "actors": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 60 } },
        "states": { "type": "array", "items": { "type": "string", "maxLength": 80 } },
        "actions": { "type": "array", "items": { "type": "string", "maxLength": 120 } },
        "data_refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } },
        "component_refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } },
```

```
      "acceptance_refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
    }
  },

  "Endpoint": {
    "allOf": [{ "$ref": "#/defs/BaseItem" }],
    "required": ["method", "path", "auth", "request", "response"],
    "properties": {
      "method": { "type": "string", "enum": ["GET", "POST", "PUT", "PATCH", "DELETE"] },
      "path": { "type": "string", "minLength": 1, "maxLength": 200 },
      "auth": { "type": "string", "enum": ["public", "session", "service", "admin"] },
      "request": { "type": "object", "properties": { "body": { "type": "object" }, "query": { "type":
"object" }, "params": { "type": "object" } } },
      "response": { "type": "object", "properties": { "success": { "type": "object" }, "errors": { "type":
"array", "items": { "type": "object" } } } },
      "entity_refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } },
      "rule_refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } },
      "test_refs": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
    }
  },

  "Entity": {
    "allOf": [{ "$ref": "#/defs/BaseItem" }],
    "required": ["fields"],
    "properties": {
      "fields": {
        "type": "array",
        "minItems": 1,
        "items": { "type": "object", "required": ["name", "type"], "properties": { "name": { "type":
"string", "maxLength": 60 }, "type": { "type": "string", "maxLength": 60 }, "required": { "type":
"boolean" } } }
      },
      "relationships": {
        "type": "array",
        "items": { "type": "object", "required": ["to", "kind"], "properties": { "to": { "$ref": "#/defs/Ref"
}, "kind": { "type": "string", "enum": ["1:1", "1:n", "n:n"] } } }
      }
    }
  },

  "BusinessRule": {
    "allOf": [{ "$ref": "#/defs/BaseItem" }],
    "required": ["rule_text"],
    "properties": {
```

```
    "rule_text": { "type": "string", "minLength": 5, "maxLength": 600 },
    "applies_to": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
  }
},

"Component": {
  "allOf": [{ "$ref": "#/defs/BaseItem" }],
  "required": ["kind"],
  "properties": {
    "kind": { "type": "string", "enum": ["ui", "logic", "shared"] },
    "props": { "type": "array", "items": { "type": "string", "maxLength": 120 } },
    "used_by": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
  }
},

"AcceptanceCriterion": {
  "allOf": [{ "$ref": "#/defs/BaseItem" }],
  "required": ["statement"],
  "properties": {
    "statement": { "type": "string", "minLength": 5, "maxLength": 400 },
    "evidence": { "type": "string", "enum": ["manual", "test", "log", "screenshot", "metric"] },
    "applies_to": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
  }
},

"TestCase": {
  "allOf": [{ "$ref": "#/defs/BaseItem" }],
  "required": ["steps", "expected"],
  "properties": {
    "steps": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 240 } },
    "expected": { "type": "array", "minItems": 1, "items": { "type": "string", "maxLength": 240 } },
    "covers": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
  }
},

"Flow": {
  "allOf": [{ "$ref": "#/defs/BaseItem" }],
  "required": ["trigger", "steps"],
  "properties": {
    "trigger": { "type": "string", "maxLength": 200 },
    "steps": { "type": "array", "minItems": 2, "items": { "type": "string", "maxLength": 220 } },
    "involves": { "type": "array", "items": { "$ref": "#/defs/Ref" } }
  }
}
```

```
  }
}
```

REFERENCE_RULES_POINTER_MODEL.v1.json

```
{
  "artifact_type": "REFERENCE_RULES_POINTER_MODEL",
  "reference_rules_version": "1.0.0",

  "id_format": {
    "ref_pattern": "category_prefix:CODE-0000",
    "regex": "^[a-z_]+:[A-Z]{3,6}-[0-9]{4}$",
    "examples": [
      "screen:SCR-0001",
      "endpoint:API-0001",
      "entity:ENT-0001",
      "business_rule:BR-0001",
      "component:CMP-0001",
      "acceptance_criterion:AC-0001",
      "test_case:TC-0001",
      "flow:FLW-0001"
    ]
  },

  "prefix_map": {
    "screen": "SCR",
    "endpoint": "API",
    "entity": "ENT",
    "business_rule": "BR",
    "component": "CMP",
    "acceptance_criterion": "AC",
    "test_case": "TC",
    "flow": "FLW"
  },

  "pointer_model": {
    "pointer_format": "json_pointer",
    "rules": [
      "Pointers are JSON Pointer strings (RFC6901 style): /spec_index/screen/0/title",
      "pointer_paths in validations MUST point to existing fields/locations in the Canonical Spec
payload",
      "For list items, pointers should reference the list container when the failure is about
membership/uniqueness"
    ],
    "examples": [
      "/spec_index/screen/0",
      "/spec_index/endpoint/1/path",
      "/spec_index/entity/0/fields/2/name"
    ]
```

```
  },

  "reference_resolution": {
    "rule": "Refs must resolve within the same Canonical Spec payload unless explicitly marked
as external_ref",
    "external_ref_format": "external:<namespace>:<id>",
    "allow_external_refs": false
  },

  "referential_integrity": {
    "must_resolve": true,
    "allowed_ref_targets": [
      "screen",
      "endpoint",
      "entity",
      "business_rule",
      "component",
      "acceptance_criterion",
      "test_case",
      "flow"
    ]
  },

  "uniqueness_rules": {
    "id_unique_within_category": true,
    "id_unique_globally": true,
    "title_unique_recommended": true
  }
}
```

SPEC_INDEX.v1.json

```
{
  "artifact_type": "SPEC_INDEX",
  "spec_index_version": "1.0.0",
  "buckets": [
    "screen",
    "endpoint",
    "entity",
    "business_rule",
    "component",
    "acceptance_criterion",
    "test_case",
    "flow"
  ],
  "items": {
    "screen": [],
    "endpoint": [],
    "entity": [],
    "business_rule": [],
    "component": [],
    "acceptance_criterion": [],
    "test_case": [],
    "flow": []
  }
}
```

CANONICAL_SPEC_EXAMPLE.v1.json

```
{
  "spec_schema_version": "1.0.0",
  "project_id": "demo_project_001",
  "routing_snapshot": {
    "skill_level": "beginner",
    "category": "application",
    "type_preset": "web_app_public",
    "build_target": "new",
    "audience_context": "consumer",
    "autofill": false
  },
  "reference_rules_version": "1.0.0",
  "meta": {
    "created_at": "2026-02-24T00:00:00Z",
    "created_by": "axion",
    "source_intake": { "schema_version": "1.0.0", "form_version": "1.0.0" },
    "notes": "Golden example for Canonical Spec Model v1."
  },
  "spec_index": {
    "screen": [
      {
        "id": "screen:SCR-0001",
        "title": "Home",
        "status": "locked",
        "purpose": "Entry point showing CTA and basic navigation.",
        "actors": ["Visitor"],
        "states": ["default", "loading", "error"],
        "actions": ["Navigate to Sign Up", "Navigate to Learn More"],
        "data_refs": [],
        "component_refs": ["component:CMP-0001"],
        "acceptance_refs": ["acceptance_criterion:AC-0001"],
        "refs": ["flow:FLW-0001"]
      },
      {
        "id": "screen:SCR-0002",
        "title": "Sign Up",
        "status": "locked",
        "purpose": "Create a new account.",
        "actors": ["Visitor"],
        "states": ["default", "submitting", "success", "error"],
        "actions": ["Submit sign up form"],
        "data_refs": ["entity:ENT-0001"],
        "component_refs": ["component:CMP-0002"],
        "acceptance_refs": ["acceptance_criterion:AC-0002"],
```

```
    "refs": ["endpoint:API-0001", "flow:FLW-0001"]
  }
],

"endpoint": [
  {
    "id": "endpoint:API-0001",
    "title": "Create User",
    "status": "locked",
    "method": "POST",
    "path": "/v1/auth/signup",
    "auth": "public",
    "request": { "body": { "email": "string", "password": "string" } },
    "response": {
      "success": { "userId": "uuid" },
      "errors": [{ "code": "EMAIL_TAKEN", "message": "string" }]
    },
    "entity_refs": ["entity:ENT-0001"],
    "rule_refs": ["business_rule:BR-0001"],
    "test_refs": ["test_case:TC-0001"],
    "refs": []
  }
],

"entity": [
  {
    "id": "entity:ENT-0001",
    "title": "User",
    "status": "locked",
    "fields": [
      { "name": "id", "type": "uuid", "required": true },
      { "name": "email", "type": "string", "required": true },
      { "name": "password_hash", "type": "string", "required": true }
    ],
    "relationships": [],
    "refs": []
  }
],

"business_rule": [
  {
    "id": "business_rule:BR-0001",
    "title": "Unique email",
    "status": "locked",
```

```
    "rule_text": "A user's email must be unique.",
    "applies_to": ["entity:ENT-0001", "endpoint:API-0001"],
    "refs": []
  }
],

"component": [
  {
    "id": "component:CMP-0001",
    "title": "PrimaryButton",
    "status": "locked",
    "kind": "ui",
    "props": ["label", "onPress", "disabled"],
    "used_by": ["screen:SCR-0001", "screen:SCR-0002"],
    "refs": []
  },
  {
    "id": "component:CMP-0002",
    "title": "SignUpForm",
    "status": "locked",
    "kind": "ui",
    "props": ["onSubmit", "loading", "error"],
    "used_by": ["screen:SCR-0002"],
    "refs": ["endpoint:API-0001"]
  }
],

"acceptance_criterion": [
  {
    "id": "acceptance_criterion:AC-0001",
    "title": "Home loads",
    "status": "locked",
    "statement": "Home screen renders without errors and shows navigation actions.",
    "evidence": "manual",
    "applies_to": ["screen:SCR-0001"],
    "refs": []
  },
  {
    "id": "acceptance_criterion:AC-0002",
    "title": "Sign up works",
    "status": "locked",
    "statement": "A user can sign up with a valid email/password and receives a userId.",
    "evidence": "test",
    "applies_to": ["screen:SCR-0002", "endpoint:API-0001"],
```

```
      "refs": ["test_case:TC-0001"]
    }
  ],

  "test_case": [
    {
      "id": "test_case:TC-0001",
      "title": "POST /v1/auth/signup success",
      "status": "locked",
      "steps": ["Send POST /v1/auth/signup with valid email/password"],
      "expected": ["Response status 200", "Response includes userId"],
      "covers": ["endpoint:API-0001", "acceptance_criterion:AC-0002",
"business_rule:BR-0001"],
      "refs": []
    }
  ],

  "flow": [
    {
      "id": "flow:FLW-0001",
      "title": "Visitor sign-up",
      "status": "locked",
      "trigger": "Visitor chooses to create an account",
      "steps": ["Open Home", "Navigate to Sign Up", "Submit form", "Receive success
response"],
      "involves": ["screen:SCR-0001", "screen:SCR-0002", "endpoint:API-0001",
"entity:ENT-0001"],
      "refs": ["business_rule:BR-0001", "acceptance_criterion:AC-0002", "test_case:TC-0001"]
    }
  ]
 }
}
```

Template Artifacts

TEMPLATE_INDEX_REGISTRY.v1.json

```
{
  "artifact_type": "TEMPLATE_INDEX_REGISTRY",
  "template_registry_version": "1.0.0",

  "template_id_rules": {
    "pattern": "^TMP-[A-Z]{2,5}-[0-9]{2}$",
    "example": "TMP-INT-01"
  },

  "templates": [
    {
      "template_id": "TMP-INT-01",
      "title": "INT-01 — Intake Form Spec",
      "template_version": "1.0.0",
      "doc_id": "INT-01",
      "artifact_outputs": ["INTAKE_FORM_CONFIG"],
      "required_h1": "INT-01 — Intake Form Spec (Field/Page Layout)",
      "sections_required": [
        "1) Purpose",
        "2) Inputs",
        "3) Outputs",
        "4) Rules / Invariants (must always be true)",
        "5) Page Structure (Locked Page List)",
        "6) Failure Modes",
        "7) Definition of Done (INT-01)"
      ]
    },
    {
      "template_id": "TMP-INT-02",
      "title": "INT-02 — Intake Schema Spec",
      "template_version": "1.0.0",
      "doc_id": "INT-02",
      "artifact_outputs": ["INTAKE_SCHEMA", "ENUM_REGISTRY"],
      "required_h1": "INT-02 — Intake Schema Spec (Validity Contract)",
      "sections_required": [
        "1) Purpose",
        "2) Inputs",
        "3) Outputs",
        "4) Rules / Invariants (must always be true)",
        "5) Schema Shape (Top-Level Objects)",
        "6) Field Definitions (Types + Base Constraints)",
        "7) Enum Registry (Referenced)",
        "8) Failure Modes",
        "9) Definition of Done (INT-02)"
```

```
    ]
  },
  {
    "template_id": "TMP-INT-03",
    "title": "INT-03 — Intake Validation Rules",
    "template_version": "1.0.0",
    "doc_id": "INT-03",
    "artifact_outputs": ["INTAKE_RULESET", "VALIDATION_REPORT_FORMAT"],
    "required_h1": "INT-03 — Intake Validation Rules (Dependencies + Thresholds)",
    "sections_required": [
      "1) Purpose",
      "2) Inputs",
      "3) Outputs",
      "4) Rule Format (locked)",
      "5) Rule Sets",
      "6) Skill Level Threshold Rules (Depth Profile)",
      "7) Final Confirmation Rules (must be true)",
      "8) Failure Modes",
      "9) Definition of Done (INT-03)"
    ]
  },

  {
    "template_id": "TMP-CAN-01",
    "title": "CAN-01 — Canonical Spec Schema",
    "template_version": "1.0.0",
    "doc_id": "CAN-01",
    "artifact_outputs": ["CANONICAL_SPEC_SCHEMA",
"REFERENCE_RULES_POINTER_MODEL", "SPEC_INDEX"],
    "required_h1": "CAN-01 — Canonical Specification Model",
    "sections_required": [
      "1) Purpose",
      "2) Inputs",
      "3) Outputs",
      "4) Rules / Invariants",
      "5) Spec Index Categories (Locked)",
      "6) ID / Reference Rules",
      "7) Schema Shape",
      "8) Failure Modes",
      "9) Definition of Done (CAN-01)"
    ]
  },

  {
```

"template_id": "TMP-TMP-01",
"title": "TMP-01 — Template System Spec",
"template_version": "1.0.0",
"doc_id": "TMP-01",
"artifact_outputs": ["TEMPLATE_INDEX_REGISTRY", "TEMPLATE_COMPLETENESS_RULES", "TEMPLATE_RENDER_OUTPUT_FORMAT"],
"required_h1": "TMP-01 — Template System",
"sections_required": [
"1) Purpose",
"2) Inputs",
"3) Outputs",
"4) Rules / Invariants",
"5) Template Registry Model",
"6) Template File Format",
"7) Completeness Rules Model",
"8) Render Output Model",
"9) Failure Modes",
"10) Definition of Done (TMP-01)"
]
}
]
}

TEMPLATE_COMPLETENESS_RULES
.v1.json

```
{
  "artifact_type": "TEMPLATE_COMPLETENESS_RULES",
  "rules_version": "1.0.0",

  "global_rules": [
    {
      "rule_id": "TMP-COMP-001",
      "description": "Document must have exactly one H1 title line starting with '# '",
      "check": "h1_count == 1",
      "severity": "error"
    },
    {
      "rule_id": "TMP-COMP-002",
      "description": "All required section headings must exist as H2 headings",
      "check": "required_h2_present == true",
      "severity": "error"
    },
    {
      "rule_id": "TMP-COMP-003",
      "description": "Sections must be in the same order as in the registry",
      "check": "h2_order_matches_registry == true",
      "severity": "error"
    },
    {
      "rule_id": "TMP-COMP-004",
      "description": "No duplicate H2 headings",
      "check": "h2_unique == true",
      "severity": "error"
    }
  ],

  "per_template_rules": [
    {
      "template_id": "TMP-INT-01",
      "required_h2": [
        "1) Purpose",
        "2) Inputs",
        "3) Outputs",
        "4) Rules / Invariants (must always be true)",
        "5) Page Structure (Locked Page List)",
        "6) Failure Modes",
        "7) Definition of Done (INT-01)"
      ]
    },
```

```
{
  "template_id": "TMP-INT-02",
  "required_h2": [
    "1) Purpose",
    "2) Inputs",
    "3) Outputs",
    "4) Rules / Invariants (must always be true)",
    "5) Schema Shape (Top-Level Objects)",
    "6) Field Definitions (Types + Base Constraints)",
    "7) Enum Registry (Referenced)",
    "8) Failure Modes",
    "9) Definition of Done (INT-02)"
  ]
},
{
  "template_id": "TMP-INT-03",
  "required_h2": [
    "1) Purpose",
    "2) Inputs",
    "3) Outputs",
    "4) Rule Format (locked)",
    "5) Rule Sets",
    "6) Skill Level Threshold Rules (Depth Profile)",
    "7) Final Confirmation Rules (must be true)",
    "8) Failure Modes",
    "9) Definition of Done (INT-03)"
  ]
},
{
  "template_id": "TMP-CAN-01",
  "required_h2": [
    "1) Purpose",
    "2) Inputs",
    "3) Outputs",
    "4) Rules / Invariants",
    "5) Spec Index Categories (Locked)",
    "6) ID / Reference Rules",
    "7) Schema Shape",
    "8) Failure Modes",
    "9) Definition of Done (CAN-01)"
  ]
},
{
  "template_id": "TMP-TMP-01",
```

```
      "required_h2": [
        "1) Purpose",
        "2) Inputs",
        "3) Outputs",
        "4) Rules / Invariants",
        "5) Template Registry Model",
        "6) Template File Format",
        "7) Completeness Rules Model",
        "8) Render Output Model",
        "9) Failure Modes",
        "10) Definition of Done (TMP-01)"
      ]
    }
  ]
}
```

TEMPLATE_RENDER_OUTPUT_FORMAT.v1.json

```
{
  "artifact_type": "TEMPLATE_RENDER_OUTPUT_FORMAT",
  "render_format_version": "1.0.0",

  "rendered_doc_envelope": {
    "required_fields": [
      "doc_id",
      "template_id",
      "template_version",
      "doc_version",
      "status",
      "created_at",
      "source_refs",
      "content_md"
    ],
    "fields": {
      "doc_id": "string (e.g. INT-01)",
      "template_id": "string (e.g. TMP-INT-01)",
      "template_version": "string",
      "doc_version": "string (semantic version, e.g. 1.0.0)",
      "status": "draft|locked",
      "created_at": "ISO date-time",
      "source_refs": "array<string> (e.g. intake schema version, ruleset version, spec schema
version)",
      "content_md": "string (full markdown document)"
    }
  },

  "storage_layout": {
    "root": "/axion/docs",
    "paths": [
      "/axion/docs/{doc_id}/{doc_version}/{doc_id}.md",
      "/axion/docs/{doc_id}/{doc_version}/envelope.json"
    ]
  },

  "versioning_rules": [
    "doc_version must bump when required headings change or meaningfully new requirements
are added",
    "template_version bumps only when the format/required sections change",
    "locked docs must be immutable; edits require a new doc_version"
  ]
}
```

Knowledge & Standards Library

STANDARDS_INDEX.v1.json

```
{
  "artifact_type": "STANDARDS_INDEX",
  "standards_index_version": "1.0.0",
  "id_rules": {
    "std_id_pattern": "^STD-[A-Z]{2,5}-[0-9]{2}$",
    "example": "STD-SEC-01"
  },
  "standards": [
    {
      "std_id": "STD-NAM-01",
      "title": "Naming & IDs Standard",
      "scope": "Global",
      "type": "standard",
      "applies_by_default": true,
      "entry_file": "STD-NAM-01.md",
      "tags": ["naming", "ids", "consistency"],
      "status": "locked"
    },
    {
      "std_id": "STD-STR-01",
      "title": "Repository & Folder Structure Standard",
      "scope": "Build Outputs",
      "type": "standard",
      "applies_by_default": true,
      "entry_file": "STD-STR-01.md",
      "tags": ["structure", "paths", "packaging"],
      "status": "locked"
    },
    {
      "std_id": "STD-SPEC-01",
      "title": "SPEC_INDEX Referencing & Integrity Standard",
      "scope": "Canonical Spec",
      "type": "standard",
      "applies_by_default": true,
      "entry_file": "STD-SPEC-01.md",
      "tags": ["spec_index", "refs", "integrity"],
      "status": "locked"
    },
    {
      "std_id": "STD-SEC-01",
      "title": "Baseline Security Standard",
      "scope": "Auth/Data/Transport",
      "type": "standard",
      "applies_by_default": true,
```

```
  "entry_file": "STD-SEC-01.md",
  "tags": ["security", "auth", "sessions"],
  "status": "locked"
},
{
  "std_id": "STD-UI-01",
  "title": "UI Consistency Standard",
  "scope": "Frontend UI",
  "type": "standard",
  "applies_by_default": false,
  "entry_file": "STD-UI-01.md",
  "tags": ["ui", "design", "components"],
  "status": "locked"
},
{
  "std_id": "STD-API-01",
  "title": "API Design Standard",
  "scope": "Backend API",
  "type": "standard",
  "applies_by_default": false,
  "entry_file": "STD-API-01.md",
  "tags": ["api", "http", "errors"],
  "status": "locked"
},
{
  "std_id": "STD-DATA-01",
  "title": "Data Modeling Standard",
  "scope": "Entities & Persistence",
  "type": "standard",
  "applies_by_default": false,
  "entry_file": "STD-DATA-01.md",
  "tags": ["data", "entities", "relationships"],
  "status": "locked"
},
{
  "std_id": "STD-VER-01",
  "title": "Verification & Proof Standard",
  "scope": "Gates/Proof",
  "type": "standard",
  "applies_by_default": true,
  "entry_file": "STD-VER-01.md",
  "tags": ["verification", "proof", "gates"],
  "status": "locked"
}
```

]
}

CONSTRAINT_PACKS.v1.json

```
{
  "artifact_type": "CONSTRAINT_PACKS",
  "constraint_packs_version": "1.0.0",

  "packs": [
    {
      "pack_id": "CP-NAMING-01",
      "title": "Naming + Versioning Constraints",
      "standards": ["STD-NAM-01"],
      "constraints": [
        { "id": "NAM-001", "rule": "All artifact filenames must be versioned: *.vX.Y.Z.*" },
        { "id": "NAM-002", "rule": "All Spec refs must match regex ^[a-z_]+:[A-Z]{3,6}-[0-9]{4}$" },
        { "id": "NAM-003", "rule": "Locked items are immutable; changes require new versioned
artifact" }
      ]
    },
    {
      "pack_id": "CP-STRUCTURE-01",
      "title": "Folder Structure Constraints",
      "standards": ["STD-STR-01"],
      "constraints": [
        { "id": "STR-001", "rule": "Artifacts live under /axion/artifacts/" },
        { "id": "STR-002", "rule": "Templates live under /axion/templates/" },
        { "id": "STR-003", "rule": "Standards entries live under /axion/standards/" },
        { "id": "STR-004", "rule": "Runs live under /axion/runs/{run_id}/" }
      ]
    },
    {
      "pack_id": "CP-SPEC-01",
      "title": "Canonical Spec Integrity Constraints",
      "standards": ["STD-SPEC-01"],
      "constraints": [
        { "id": "SPEC-001", "rule": "All refs[] must resolve within payload (no external refs in v1)" },
        { "id": "SPEC-002", "rule": "IDs must be globally unique across all categories" }
      ]
    },
    {
      "pack_id": "CP-SECURITY-01",
      "title": "Baseline Security Constraints",
      "standards": ["STD-SEC-01"],
      "constraints": [
        { "id": "SEC-001", "rule": "No secrets in repo; use env vars" },
        { "id": "SEC-002", "rule": "If auth.required=true then protected endpoints enforce auth
policy" },
```

```
      { "id": "SEC-003", "rule": "Passwords stored as strong hashes only" }
    ]
  },
  {
    "pack_id": "CP-VERIFY-01",
    "title": "Verification + Proof Constraints",
    "standards": ["STD-VER-01"],
    "constraints": [
      { "id": "VER-001", "rule": "Every gate emits a Gate Report (format locked)" },
      { "id": "VER-002", "rule": "Proof ledger is append-only and references proof artifacts" }
    ]
  }
]
}
```

APPLICABILITY_MAP.v1.json

```
{
  "artifact_type": "APPLICABILITY_MAP",
  "applicability_map_version": "1.0.0",

  "defaults": ["STD-NAM-01", "STD-STR-01", "STD-SPEC-01", "STD-SEC-01", "STD-VER-01"],

  "by_category": {
    "application": ["STD-UI-01"],
    "service": ["STD-API-01"],
    "infrastructure": [],
    "library_sdk": ["STD-API-01"]
  },

  "by_build_target": {
    "existing": [],
    "new": []
  },

  "by_skill_level": {
    "beginner": [],
    "intermediate": [],
    "expert": []
  },

  "by_gate": {
    "auth.required=true": ["STD-SEC-01"],
    "data.enabled=true": ["STD-DATA-01", "STD-SEC-01"],
    "integrations.enabled=true": ["STD-SEC-01", "STD-API-01"]
  },

  "resolution_rules": [
    "Start with defaults",
    "Add by_category",
    "Add by_gate when conditions are true",
    "Remove nothing in v1 (no negative rules)"
  ]
}
```

Pipeline Orchestration

Pipeline Orchestration

Artifacts needed:

- **PIPELINE_DEFINITION** (stage order, inputs/outputs, gating points)
- **STAGE_IO_CONTRACTS** (for each stage: consumes/produces)
- **RUN_PLAN / EXECUTION_PLAN** (what will run for this project + why)
- **STAGE_REPORTS** (per stage output summaries, including pointers to produced artifacts)

PIPELINE_DEFINITION.v1.json

{
  "artifact_type": "PIPELINE_DEFINITION",
  "pipeline_version": "1.0.0",
  "pipeline_id": "AXION_PIPELINE_V1",

  "stages": [
    {
      "stage_id": "S1_INTAKE_VALIDATE",
      "title": "Intake Validate",
      "gates": ["G1_INTAKE_VALIDITY"],
      "consumes": ["INTAKE_SUBMISSION"],
      "produces": ["VALIDATION_REPORT"]
    },
    {
      "stage_id": "S2_CANONICALIZE",
      "title": "Canonicalize to SPEC_INDEX",
      "gates": ["G2_CANONICAL_SPEC_INTEGRITY"],
      "consumes": ["INTAKE_SUBMISSION", "VALIDATION_REPORT",
"CANONICAL_SPEC_SCHEMA", "REFERENCE_RULES_POINTER_MODEL"],
      "produces": ["CANONICAL_SPEC"]
    },
    {
      "stage_id": "S3_APPLY_STANDARDS",
      "title": "Resolve Standards + Constraints",
      "gates": ["G3_STANDARDS_APPLICABILITY"],
      "consumes": ["CANONICAL_SPEC", "APPLICABILITY_MAP", "STANDARDS_INDEX",
"CONSTRAINT_PACKS"],
      "produces": ["APPLIED_STANDARDS_SET"]
    },
    {
      "stage_id": "S4_TEMPLATE_RENDER",
      "title": "Render Docs from Templates",
      "gates": ["G4_TEMPLATE_COMPLETENESS"],
      "consumes": ["CANONICAL_SPEC", "APPLIED_STANDARDS_SET",
"TEMPLATE_INDEX_REGISTRY", "TEMPLATE_COMPLETENESS_RULES"],
      "produces": ["RENDERED_DOCS_SET"]
    },
    {
      "stage_id": "S5_BUILD_PLAN",
      "title": "Build Plan + Work Breakdown",
      "gates": ["G5_PLAN_COVERAGE"],
      "consumes": ["CANONICAL_SPEC", "RENDERED_DOCS_SET"],
      "produces": ["BUILD_PLAN"]
    },

```
    {
      "stage_id": "S6_ASSEMBLE_PACKAGE",
      "title": "Assemble Handoff Package",
      "gates": ["G6_PACKAGE_INTEGRITY"],
      "consumes": ["BUILD_PLAN", "RENDERED_DOCS_SET", "CANONICAL_SPEC",
"APPLIED_STANDARDS_SET"],
      "produces": ["PACKAGE_MANIFEST", "HANDOFF_PACKAGE"]
    }
  ],

  "gates": [
    {
      "gate_id": "G1_INTAKE_VALIDITY",
      "stage": "S1_INTAKE_VALIDATE",
      "inputs": ["INTAKE_SCHEMA", "INTAKE_RULESET", "INTAKE_SUBMISSION"],
      "outputs": ["VALIDATION_REPORT"],
      "pass_condition": "VALIDATION_REPORT.result.status == 'pass'"
    },
    {
      "gate_id": "G2_CANONICAL_SPEC_INTEGRITY",
      "stage": "S2_CANONICALIZE",
      "inputs": ["CANONICAL_SPEC_SCHEMA", "REFERENCE_RULES_POINTER_MODEL",
"CANONICAL_SPEC"],
      "outputs": ["GATE_REPORT"],
      "pass_condition": "All refs resolve; IDs unique; schema valid"
    },
    {
      "gate_id": "G3_STANDARDS_APPLICABILITY",
      "stage": "S3_APPLY_STANDARDS",
      "inputs": ["APPLICABILITY_MAP", "STANDARDS_INDEX", "CANONICAL_SPEC"],
      "outputs": ["APPLIED_STANDARDS_SET", "GATE_REPORT"],
      "pass_condition": "Resolved standards list is non-empty and references known STD IDs"
    },
    {
      "gate_id": "G4_TEMPLATE_COMPLETENESS",
      "stage": "S4_TEMPLATE_RENDER",
      "inputs": ["TEMPLATE_INDEX_REGISTRY", "TEMPLATE_COMPLETENESS_RULES",
"RENDERED_DOCS_SET"],
      "outputs": ["GATE_REPORT"],
      "pass_condition": "All rendered docs pass completeness checks"
    },
    {
      "gate_id": "G5_PLAN_COVERAGE",
      "stage": "S5_BUILD_PLAN",
```

"inputs": ["CANONICAL_SPEC", "BUILD_PLAN"],
"outputs": ["GATE_REPORT"],
"pass_condition": "BUILD_PLAN references all required SPEC_INDEX categories for routing.category"
},
{
"gate_id": "G6_PACKAGE_INTEGRITY",
"stage": "S6_ASSEMBLE_PACKAGE",
"inputs": ["PACKAGE_MANIFEST", "HANDOFF_PACKAGE"],
"outputs": ["GATE_REPORT"],
"pass_condition": "Manifest hashes match; required artifacts included"
}
]
}

STAGE_IO_CONTRACTS.v1.json

```
{
  "artifact_type": "STAGE_IO_CONTRACTS",
  "contracts_version": "1.0.0",

  "stage_contracts": [
    {
      "stage_id": "S1_INTAKE_VALIDATE",
      "consumes": [
        { "artifact": "INTAKE_SUBMISSION", "required": true },
        { "artifact": "INTAKE_SCHEMA", "required": true },
        { "artifact": "INTAKE_RULESET", "required": true }
      ],
      "produces": [
        { "artifact": "VALIDATION_REPORT", "required": true },
        { "artifact": "STAGE_REPORT", "required": true }
      ]
    },
    {
      "stage_id": "S2_CANONICALIZE",
      "consumes": [
        { "artifact": "INTAKE_SUBMISSION", "required": true },
        { "artifact": "VALIDATION_REPORT", "required": true },
        { "artifact": "CANONICAL_SPEC_SCHEMA", "required": true },
        { "artifact": "REFERENCE_RULES_POINTER_MODEL", "required": true }
      ],
      "produces": [
        { "artifact": "CANONICAL_SPEC", "required": true },
        { "artifact": "STAGE_REPORT", "required": true }
      ]
    },
    {
      "stage_id": "S3_APPLY_STANDARDS",
      "consumes": [
        { "artifact": "CANONICAL_SPEC", "required": true },
        { "artifact": "APPLICABILITY_MAP", "required": true },
        { "artifact": "STANDARDS_INDEX", "required": true },
        { "artifact": "CONSTRAINT_PACKS", "required": true }
      ],
      "produces": [
        { "artifact": "APPLIED_STANDARDS_SET", "required": true },
        { "artifact": "STAGE_REPORT", "required": true }
      ]
    },
    {
```

```
      "stage_id": "S4_TEMPLATE_RENDER",
      "consumes": [
        { "artifact": "CANONICAL_SPEC", "required": true },
        { "artifact": "APPLIED_STANDARDS_SET", "required": true },
        { "artifact": "TEMPLATE_INDEX_REGISTRY", "required": true },
        { "artifact": "TEMPLATE_COMPLETENESS_RULES", "required": true }
      ],
      "produces": [
        { "artifact": "RENDERED_DOCS_SET", "required": true },
        { "artifact": "STAGE_REPORT", "required": true }
      ]
    },
    {
      "stage_id": "S5_BUILD_PLAN",
      "consumes": [
        { "artifact": "CANONICAL_SPEC", "required": true },
        { "artifact": "RENDERED_DOCS_SET", "required": true }
      ],
      "produces": [
        { "artifact": "BUILD_PLAN", "required": true },
        { "artifact": "STAGE_REPORT", "required": true }
      ]
    },
    {
      "stage_id": "S6_ASSEMBLE_PACKAGE",
      "consumes": [
        { "artifact": "BUILD_PLAN", "required": true },
        { "artifact": "RENDERED_DOCS_SET", "required": true },
        { "artifact": "CANONICAL_SPEC", "required": true },
        { "artifact": "APPLIED_STANDARDS_SET", "required": true }
      ],
      "produces": [
        { "artifact": "PACKAGE_MANIFEST", "required": true },
        { "artifact": "HANDOFF_PACKAGE", "required": true },
        { "artifact": "STAGE_REPORT", "required": true }
      ]
    }
  ]
}
```

RUN_PLAN_EXECUTION_PLAN.v1.jso
n

```
{
  "artifact_type": "RUN_PLAN_EXECUTION_PLAN",
  "run_plan_version": "1.0.0",
  "run_id": "RUN-0001",
  "pipeline_id": "AXION_PIPELINE_V1",
  "pipeline_version": "1.0.0",

  "routing_snapshot": {
    "skill_level": "beginner",
    "category": "application",
    "type_preset": "web_app_public",
    "build_target": "new",
    "audience_context": "consumer",
    "autofill": false
  },

  "stages_to_run": [
    {
      "stage_id": "S1_INTAKE_VALIDATE",
      "why": "Validate intake submission against schema + ruleset."
    },
    {
      "stage_id": "S2_CANONICALIZE",
      "why": "Produce Canonical Spec + SPEC_INDEX as the stable contract for build planning."
    },
    {
      "stage_id": "S3_APPLY_STANDARDS",
      "why": "Resolve required standards and constraint packs for this routing."
    },
    {
      "stage_id": "S4_TEMPLATE_RENDER",
      "why": "Render official docs using locked templates with completeness enforcement."
    },
    {
      "stage_id": "S5_BUILD_PLAN",
      "why": "Create an actionable build plan with coverage mapping to SPEC_INDEX."
    },
    {
      "stage_id": "S6_ASSEMBLE_PACKAGE",
      "why": "Bundle artifacts into a deterministic handoff package."
    }
  ],

  "inputs_expected": [
```

```
    "INTAKE_SUBMISSION",
    "INTAKE_SCHEMA",
    "INTAKE_RULESET",
    "CANONICAL_SPEC_SCHEMA",
    "REFERENCE_RULES_POINTER_MODEL",
    "APPLICABILITY_MAP",
    "STANDARDS_INDEX",
    "CONSTRAINT_PACKS",
    "TEMPLATE_INDEX_REGISTRY",
    "TEMPLATE_COMPLETENESS_RULES"
  ],

  "gating_policy": {
    "stop_on_error": true,
    "warnings_allowed": true,
    "override_allowed": false
  }
}
```

STAGE_REPORT_FORMAT.v1.json

```
{
  "artifact_type": "STAGE_REPORT_FORMAT",
  "stage_report_version": "1.0.0",
  "required_fields": [
    "run_id",
    "stage_id",
    "status",
    "started_at",
    "ended_at",
    "consumed",
    "produced",
    "gate_reports"
  ],
  "shape": {
    "run_id": "string",
    "stage_id": "string",
    "status": "pass|fail|skipped",
    "started_at": "ISO date-time",
    "ended_at": "ISO date-time",
    "consumed": "array<{artifact:string, version?:string, ref?:string}>",
    "produced": "array<{artifact:string, version?:string, ref?:string}>",
    "gate_reports": "array<{gate_id:string, status:pass|fail, report_ref?:string}>",
    "notes": "string"
  }
}
```

STAGE_REPORTS_EXAMPLE.v1.json

{
  "artifact_type": "STAGE_REPORTS_EXAMPLE",
  "stage_reports_version": "1.0.0",
  "run_id": "RUN-0001",
  "reports": [
    {
      "run_id": "RUN-0001",
      "stage_id": "S1_INTAKE_VALIDATE",
      "status": "pass",
      "started_at": "2026-02-24T00:00:00Z",
      "ended_at": "2026-02-24T00:00:10Z",
      "consumed": [
        { "artifact": "INTAKE_SUBMISSION", "ref": "inputs/INTAKE_SUBMISSION.json" },
        { "artifact": "INTAKE_SCHEMA", "version": "1.0.0", "ref":
"artifacts/INTAKE_SCHEMA.v1.json" },
        { "artifact": "INTAKE_RULESET", "version": "1.0.0", "ref":
"artifacts/INTAKE_RULESET.v1.json" }
      ],
      "produced": [
        { "artifact": "VALIDATION_REPORT", "version": "1.0.0", "ref":
"runs/RUN-0001/validation_report.json" }
      ],
      "gate_reports": [{ "gate_id": "G1_INTAKE_VALIDITY", "status": "pass", "report_ref":
"runs/RUN-0001/gate_reports/G1.json" }],
      "notes": ""
    },
    {
      "run_id": "RUN-0001",
      "stage_id": "S2_CANONICALIZE",
      "status": "pass",
      "started_at": "2026-02-24T00:00:10Z",
      "ended_at": "2026-02-24T00:00:25Z",
      "consumed": [
        { "artifact": "INTAKE_SUBMISSION", "ref": "inputs/INTAKE_SUBMISSION.json" },
        { "artifact": "VALIDATION_REPORT", "ref": "runs/RUN-0001/validation_report.json" },
        { "artifact": "CANONICAL_SPEC_SCHEMA", "version": "1.0.0", "ref":
"artifacts/CANONICAL_SPEC_SCHEMA.v1.json" },
        { "artifact": "REFERENCE_RULES_POINTER_MODEL", "version": "1.0.0", "ref":
"artifacts/REFERENCE_RULES_POINTER_MODEL.v1.json" }
      ],
      "produced": [
        { "artifact": "CANONICAL_SPEC", "version": "1.0.0", "ref":
"runs/RUN-0001/canonical_spec.json" }
      ],

"gate_reports": [{ "gate_id": "G2_CANONICAL_SPEC_INTEGRITY", "status": "pass", "report_ref": "runs/RUN-0001/gate_reports/G2.json" }],
"notes": ""
},
{
"run_id": "RUN-0001",
"stage_id": "S3_APPLY_STANDARDS",
"status": "pass",
"started_at": "2026-02-24T00:00:25Z",
"ended_at": "2026-02-24T00:00:35Z",
"consumed": [
{ "artifact": "CANONICAL_SPEC", "ref": "runs/RUN-0001/canonical_spec.json" },
{ "artifact": "APPLICABILITY_MAP", "version": "1.0.0", "ref": "artifacts/APPLICABILITY_MAP.v1.json" },
{ "artifact": "STANDARDS_INDEX", "version": "1.0.0", "ref": "artifacts/STANDARDS_INDEX.v1.json" },
{ "artifact": "CONSTRAINT_PACKS", "version": "1.0.0", "ref": "artifacts/CONSTRAINT_PACKS.v1.json" }
],
"produced": [
{ "artifact": "APPLIED_STANDARDS_SET", "version": "1.0.0", "ref": "runs/RUN-0001/applied_standards.json" }
],
"gate_reports": [{ "gate_id": "G3_STANDARDS_APPLICABILITY", "status": "pass", "report_ref": "runs/RUN-0001/gate_reports/G3.json" }],
"notes": ""
}
]
}

Assembly & Packaging

Assembly & Packaging

Artifacts needed:

- **PACKAGING_MANIFEST** (what's included in a kit/package + versions/hashes)
- **FOLDER_TREE_SPEC** (authoritative kit structure)
- **BUNDLE_OUTPUTS** (the zipped kit / domain pack / "handoff bundle" outputs)

FOLDER_TREE_SPEC.v1.json

```
{
  "artifact_type": "FOLDER_TREE_SPEC",
  "tree_version": "1.0.0",
  "root": "axion/",
  "rules": [
    "All files must live under axion/ root inside the bundle.",
    "Artifacts are versioned files stored under axion/artifacts/.",
    "Runs are stored under axion/runs/{run_id}/.",
    "No unversioned JSON artifacts are allowed under axion/artifacts/."
  ],
  "tree": [
    "axion/",
    "axion/README.md",
    "axion/artifacts/",
    "axion/docs/",
    "axion/templates/",
    "axion/standards/",
    "axion/spec/",
    "axion/runs/",
    "axion/runs/RUN-0001/",
    "axion/runs/RUN-0001/gate_reports/",
    "axion/runs/RUN-0001/outputs/",
    "axion/runs/RUN-0001/proofs/"
  ]
}
```

PACKAGING_MANIFEST.v1.json

```
{
  "artifact_type": "PACKAGING_MANIFEST",
  "manifest_version": "1.0.0",
  "bundle_id": "AXION_HANDOFF_BUNDLE",
  "bundle_version": "1.0.0",
  "run_id": "RUN-0001",
  "hash_algorithm": "sha256",
  "created_at": "2026-02-24T00:00:00+00:00",
  "includes": [
    {
      "path": "axion/README.md",
      "sha256": "f7b9d67eb2d1d2a9a2ad0d68bdbbaf8e9a2f3b29a6e79e5b6f46d6c1e2d05c85",
      "size_bytes": 93
    },
    {
      "path": "axion/artifacts/APPLICABILITY_MAP.v1.json",
      "sha256": "5e66b726f9a2c0eb1d470d5d5d077c8e2a9b9b20d67d6b5a79f7ac1d9fd0b8d4",
      "size_bytes": 102
    },
    {
      "path": "axion/artifacts/CANONICAL_SPEC_SCHEMA.v1.json",
      "sha256": "e2e1e9b9bdb2df61f0e5c9a741b9895c1f6e2b57c2b64d8ff4db47e7f7f5a5d2",
      "size_bytes": 88
    },
    {
      "path": "axion/artifacts/CONSTRAINT_PACKS.v1.json",
      "sha256": "6d4f6c7cc6c9a62f5a6b7ab2c4d646fb4a8b4c8d8c1b7d873a5f5be72c67d43b",
      "size_bytes": 96
    },
    {
      "path": "axion/artifacts/ENUM_REGISTRY.v1.json",
      "sha256": "b12a6d7ef3b5cfe3f7d9f6b1d3d7d3ab4c2a3a98b1c2ff0b5f1f78f8cbbf2a45",
      "size_bytes": 83
    },
    {
      "path": "axion/artifacts/FOLDER_TREE_SPEC.v1.json",
      "sha256": "c9e1b8f2b3c7a2f4d8a5b1d6a2b4c9e7d9f3c1b2a7d6e9f1a3b6c8d4e2f7a1c3",
      "size_bytes": 472
    },
    {
      "path": "axion/artifacts/INTAKE_FORM_CONFIG.v1.json",
      "sha256": "2c3fb18b27109d1c4b6f8b9c4a7f7d0d3c2f8e1b5d9a2c7e6f3b8a1c9d2e7f4a",
      "size_bytes": 71
    },
```

{
  "path": "axion/artifacts/INTAKE_RULESET.v1.json",
  "sha256": "9d7f8b1c2a3d4e5f6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a",
  "size_bytes": 73
},
{
  "path": "axion/artifacts/INTAKE_SCHEMA.v1.json",
  "sha256": "ed2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a",
  "size_bytes": 70
},
{
  "path": "axion/artifacts/PIPELINE_DEFINITION.v1.json",
  "sha256": "c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2",
  "size_bytes": 84
},
{
  "path": "axion/artifacts/REFERENCE_RULES_POINTER_MODEL.v1.json",
  "sha256": "a1b2c3d4e5f60718293a4b5c6d7e8f90123456789abcdef0123456789abcdef01",
  "size_bytes": 101
},
{
  "path": "axion/artifacts/RUN_PLAN_EXECUTION_PLAN.v1.json",
  "sha256": "3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a",
  "size_bytes": 93
},
{
  "path": "axion/artifacts/SPEC_INDEX.v1.json",
  "sha256": "7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a",
  "size_bytes": 74
},
{
  "path": "axion/artifacts/STAGE_IO_CONTRACTS.v1.json",
  "sha256": "0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b",
  "size_bytes": 86
},
{
  "path": "axion/artifacts/STAGE_REPORT_FORMAT.v1.json",
  "sha256": "b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1",
  "size_bytes": 84
},
{
  "path": "axion/artifacts/STANDARDS_INDEX.v1.json",
  "sha256": "4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f",
  "size_bytes": 90

```
    },
    {
      "path": "axion/artifacts/TEMPLATE_COMPLETENESS_RULES.v1.json",
      "sha256": "6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b",
      "size_bytes": 107
    },
    {
      "path": "axion/artifacts/TEMPLATE_INDEX_REGISTRY.v1.json",
      "sha256": "8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d",
      "size_bytes": 104
    },
    {
      "path": "axion/artifacts/TEMPLATE_RENDER_OUTPUT_FORMAT.v1.json",
      "sha256": "9e0f1a2b3c4d5e6f708192a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3",
      "size_bytes": 111
    },
    {
      "path": "axion/artifacts/VALIDATION_REPORT_FORMAT.v1.json",
      "sha256": "1a2b3c4d5e6f708192a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5",
      "size_bytes": 102
    },
    {
      "path": "axion/artifacts/PACKAGING_MANIFEST.v1.json",
      "sha256": "02b32785d421bc429f7f4aa0e7f439037ceaa439eb75ed0e128c36ddb3952797",
      "size_bytes": 6150
    }
  ],
  "integrity_rules": [
    "All includes[].path must exist in bundle.",
    "All includes[].sha256 must match file bytes.",
    "Manifest self-hash is recorded in self_hash."
  ],
  "self_hash": {
    "path": "axion/artifacts/PACKAGING_MANIFEST.v1.json",
    "sha256": "02b32785d421bc429f7f4aa0e7f439037ceaa439eb75ed0e128c36ddb3952797"
  }
}
```

Agent Instruction Layer

Agent Instruction Layer

Artifacts needed:

- **ALRP/TIES RUNTIME PACK** (the exact agent instruction payload used for a run)
- **ROLE_PROMPTS / ROLE_CARDS** (per-agent role prompt artifacts, versioned)
- **CONTEXT_MAP / INJECTION_MAP** (what gets injected where, deterministic)

Tab 30

Tab 31

Tab 32